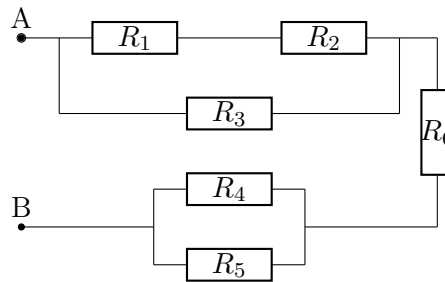
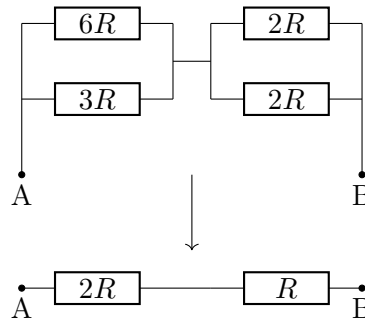


Exercise 1: Networks of resistors

- a) The resistors R_1 and R_2 are in series so that their equivalent resistance is given by $R' = R_1 + R_2 = 2R$. In a similar fashion we connect the resistors R' and R_3 which are in parallel: $(1/R'') = (1/R') + (1/R_3) = (1/2R) + (1/2R) = 1/R$ in a way that $R'' = R$. In the same way we replace R_4 and R_5 by $R''' = R$. Now we are only left with three resistors R'' , R_6 and R''' which are connected in series. This gives the total resistance of $R_{tot} = R'' + R_6 + R''' = 3R$.

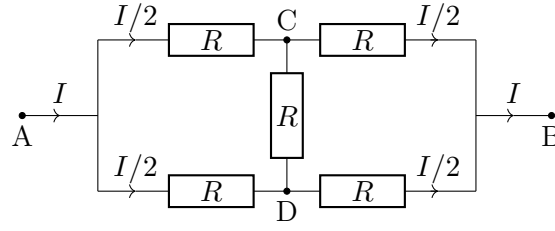


- b) This network can be rearranged like in the figure below. This way, two resistors are always connected in parallel. The two resistors on the left side combine to $R' = \left((1/6R) + (1/3R) \right)^{-1} = (1/2R)^{-1} = 2R$. The two resistors on the right side combine to $R'' = \left((1/2R) + (1/2R) \right)^{-1} = (1/R)^{-1} = R$. Finally we combine R' and R'' , which are connected in series to calculate the total equivalent resistance between A and B as $R_{tot} = R' + R'' = 3R$.



- c) The network can be simplified as shown in the drawing. The current I that enters through A is split equally between both lines for symmetry reasons. Since the current going out from B has to be I again, the current between C and D has to be zero. We neglect any instantaneous transmission of single carriers from C to D or from D to C. This is why we can ignore the part of the network between C and D.

The resistors between A and C, and C and B are connected in series and give the combined value of $R' = R + R = 2R$. In the same way the resistors between A and D, and D and B give $R'' = R + R = 2R$. R' and R'' are now connected in parallel and add to the wanted equivalent resistance $R_{tot} = \left((1/2R) + (1/2R) \right)^{-1} = (1/R)^{-1} = R$.



Exercise 2: Practical examples

- a) The resistance of the wire at room temperature is

$$R_{20} = \rho \frac{l}{\pi(d/2)^2} = 65.7 \Omega. \quad (1)$$

At operation temperature we find

$$R_{2700} = R_{20}(1 + \alpha T) = R_{20}(1 + \alpha(\vartheta - 20)) = 858 \Omega. \quad (2)$$

- b) Using Ohm's law we find the current directly after turning on and during normal operation:

$$I_{20} = \frac{V}{R_{20}} = 3.5 \text{ A}, \quad I_{2700} = \frac{V}{R_{2700}} = 0.268 \text{ A}. \quad (3)$$

You can see that the current directly after turning on is 13 times higher. This is when light bulbs are most likely to break.

- c) The power of the emitted light is given by the fraction of electrical power left over after multiplying with the efficiency

$$P = V I_{2700} \eta = 62 \text{ W} \cdot 0.05 = 3.1 \text{ W}. \quad (4)$$

Modern light-emitting diodes based on semiconductors have a 5 times higher efficiency. The other power is dissipated in longer-wavelength radiation i.e. heat.

- d) If we use equations (4.48) and (4.50) from the script we get 2 equations for the two unknown resistance values.

$$I = \frac{V_0}{R_a + R_i}, \quad V_1 = V_0 \frac{1}{1 + \frac{R_i}{R_a}}. \quad (5)$$

These can be solved for R_a and R_i , resulting in

$$R_a = \frac{V_1}{I} = 0.067 \Omega, \quad R_i = \frac{V_0 - V_1}{I} = 0.013 \Omega. \quad (6)$$

You can also get this result without a long calculation simply by looking at the equivalent circuit diagram shown in Figure,2.1. In this diagram you can see an ideal battery with two resistors in series.

- e) At low temperatures we have the same value for the internal and external resistance. This means the effectively available voltage is cut in half as can be seen in equation (4.50) from the script.

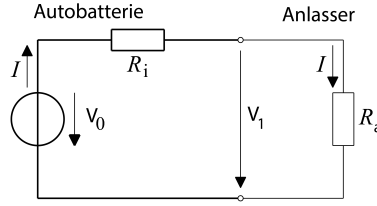


Abbildung 2.1: equivalent circuit diagram

f) The total power

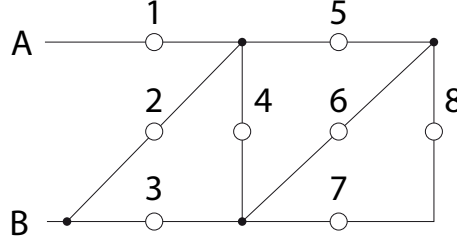
$$P = V_0 I = \frac{V_0^2}{R_i + R_a} \quad (7)$$

is split between the battery and starter depending on the individual voltage drop so that

$$P_a = V_1 I = \frac{V_0 V_1}{R_i + R_a}, \quad P_i = P - P_a = (V_0 - V_1) I = \frac{V_0 (V_0 - V_1)}{R_i + R_a}. \quad (8)$$

This gives for numbers of part a) $P_a = 1500 \text{ W}$ and $P_i = 300 \text{ W}$. In part b) on the other hand $P_a = P_i = 540 \text{ W}$.

Exercise 3: Switching circuit with capacitors



Resistors in series add up exactly as capacitors in parallel, and vice-versa:

$$R_{\text{serie}} = \sum_i R_i, \quad C_{\text{parallel}} = \sum_i C_i \quad (9)$$

and

$$\frac{1}{R_{\text{parallel}}} = \sum_i \frac{1}{R_i}, \quad \frac{1}{C_{\text{serie}}} = \sum_i \frac{1}{C_i}. \quad (10)$$

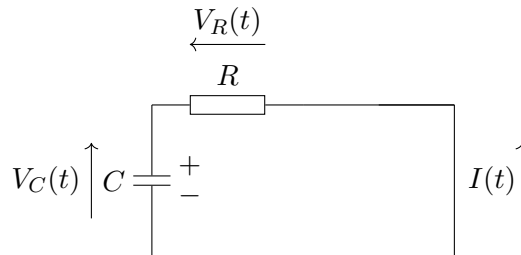
It follows immediately that the result of a) will take the exact inverse values of b). The equivalent quantity (resistance or capacitance) can be computed by splitting the circuit in several parts and by summing up:

Part	Type	C_g	R_g
7+8=a	series	$\frac{1}{2} \text{ C}$	2 R
6+a=b	parallel	$\frac{3}{2} \text{ C}$	$\frac{2}{3} \text{ R}$
5+b=c	series	$\frac{3}{5} \text{ C}$	$\frac{5}{3} \text{ R}$
4+c=d	parallel	$\frac{8}{5} \text{ C}$	$\frac{5}{8} \text{ R}$
3+d=e	series	$\frac{8}{13} \text{ C}$	$\frac{13}{8} \text{ R}$
2+e=f	parallel	$\frac{21}{13} \text{ C}$	$\frac{13}{21} \text{ R}$
1+f	series	$\frac{21}{34} \text{ C}$	$\frac{34}{21} \text{ R}$

And so the total resistance is $\frac{21}{34} \text{ C}$ and the total capacitance is $\frac{34}{21} \text{ R}$.

Exercise 4: Circuit

- a) When the switch is connected to a , a current flows on the capacitor and therefore through the lamp. As soon as the capacitor is fully charged, the current stops. Due to this process happening very fast (depending on the values of C , R and V), the lamp flashes once. The same happens when the switch is connected to b : The charge flows off the capacitor and makes the lamp flash again.
- b) To calculate the time-dependent current for $t > 0$ we have to solve the differential equation that describes the circuit with the initial conditions $V_C(0) = V$, $Q(0) = CV$.



The loop rule gives

$$RI(t) - \frac{Q(t)}{C} = 0. \quad (11)$$

As the capacitor C is discharged the following holds

$$\dot{Q} = -I, \quad (12)$$

The time-derivative of equation (11) leads to the differential equation (homogeneous, first order):

$$R\dot{I}(t) + \frac{I(t)}{C} = 0. \quad (13)$$

Insertion of the ansatz

$$I(t) = Ae^{Bt} \quad (14)$$

gives the condition

$$BR + \frac{1}{C} = 0. \quad (15)$$

Solving for B ,

$$B = -\frac{1}{RC}. \quad (16)$$

At time $t = 0$ the capacitor C has the voltage V and acts as a voltage source as soon as it is contacted with b while the initial current is $I(0) = A = V/R$. This leads to the solution:

$$I(t) = \frac{V}{R} e^{-\frac{t}{RC}}. \quad (17)$$

c) The power dissipated in the resistor is

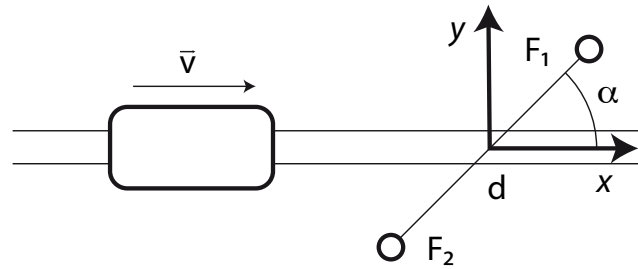
$$P_{\text{diss}}(t) = RI(t)^2 = \frac{V^2}{R} e^{-\frac{2t}{RC}}. \quad (18)$$

The integration with respect to time yields (where $t_s \approx \infty$):

$$E_{\text{diss}} = \int_0^\infty P_{\text{diss}}(t) dt = \frac{V^2}{R} \left[-\frac{RC}{2} e^{-\frac{2t}{RC}} \right]_0^\infty = \frac{1}{2} CV^2. \quad (19)$$

The dissipated energy is exactly the energy that was previously stored in the capacitor.

Exercise 5: Galilean transformations



a) We take the ground as our home reference system. The origin of the system will be the intermediate point between the two firecrackers, in such a way that the position of F1 and F2 is given by:

$$\begin{aligned} x(F1) &= d/2 \cos(\alpha) \\ y(F1) &= d/2 \sin(\alpha) \end{aligned}$$

and

$$\begin{aligned} x(F2) &= -d/2 \cos(\alpha) \\ y(F2) &= -d/2 \sin(\alpha). \end{aligned}$$

Applying the Galilean transformation in the direction x , we find in the system of reference of the train:

$$\begin{aligned}x'(F1) &= x(F1) - vt = d/2 \cos(\alpha) - vt \\y'(F1) &= y(F1) - 0 = d/2 \sin(\alpha)\end{aligned}$$

and

$$\begin{aligned}x'(F2) &= x(F2) - vt = -d/2 \cos(\alpha) - vt \\y'(F2) &= y(F2) - 0 = -d/2 \sin(\alpha).\end{aligned}$$

The positions of the firecrackers at the moment of the explosion are:

$$\begin{aligned}x'(F1) &= x(F1) - vt = d/2 \cos(\alpha) - vt \\y'(F1) &= y(F1) - 0 = d/2 \sin(\alpha)\end{aligned}$$

and

$$\begin{aligned}x'(F2) &= x(F2) - v \cdot (t + \Delta t) = -d/2 \cos(\alpha) - v \cdot (t + \Delta t) \\y'(F2) &= y(F2) - 0 = -d/2 \sin(\alpha).\end{aligned}$$

The distance is given by:

$$\begin{aligned}D(F1, F2) &= \sqrt{(x'(F1) - x'(F2))^2 + (y'(F1) - y'(F2))^2} \\&= \sqrt{(x(F1) - x(F2) + vt - vt + v\Delta t)^2 + (y(F1) - y(F2))^2} \\&= \sqrt{(d \cos(\alpha) + v\Delta t)^2 + (d \sin(\alpha))^2} \\&= \sqrt{d^2 + 2d \cos(\alpha)v\Delta t + v^2\Delta t^2}.\end{aligned}$$

In the special case where the two firecrackers explode at the same time, the distance between the explosions in the system of reference of the train is the following:

$$\begin{aligned}D(F1, F2) &= \sqrt{(x'(F1) - x'(F2))^2 + (y'(F1) - y'(F2))^2} \\&= \sqrt{(x(F1) - x(F2) + vt - vt)^2 + (y(F1) - y(F2))^2} \\&= \sqrt{(d \cos(\alpha))^2 + (d \sin(\alpha))^2} = d.\end{aligned}$$

- b) This distance is maximal when $\cos(\alpha) = 1$ and therefore the two firecrackers are aligned in the direction of the train, F_2 being closest to the train. The minimal distance occurs when $\cos(\alpha) = -1$ in such a way that the two firecrackers are again aligned but now F_1 is the one closest to the train.

This result can be understood very easily if we imagine the system of reference of the train. In that system the train is at rest and the two firecrackers are moving with velocity $-v$. When the first firecracker explodes the second one will move $-v\Delta t$ before exploding.